

SOFTWARE ENGINEERING LAB
PRACTICAL NO: 3
CLASS DIAGRAM – WAREHOUSE MANAGEMENT SYSTEM

NAME:	PRANAV RATHI
SECTION:	A1/B4
ROLL NO:	52

AIM:

To design a Class Diagram to represent the structural view of the Warehouse Management System

Study Title:

Warehouse Management System

Problem Statement:

Warehouse management can lead to stock inaccuracies, delayed order fulfillment, and poor record-keeping. The Warehouse Management System provides a simple way to track inventory, manage orders, process payments, and handle shipments.

List of Classes and Services

Class	Purpose
Supplier	Handles supplier and purchase-order operations.
Customer	Handles customer and sales-order operations.
PurchaseOrder	Stores purchase order details and status.
SalesOrder	Stores sales order details and status.
PurchaseOrderItem	Stores quantity and unit price for purchase items.
SalesOrderItem	Stores quantity and unit price for sales items.

Product	Stores product details and SKU.
InventoryItem	Stores quantity and reorder level.
Payment	Stores payment details and method.
Shipment	Stores shipment and tracking details.

UML Class Diagram

The following diagram represents the structural design of the Warehouse Management System.

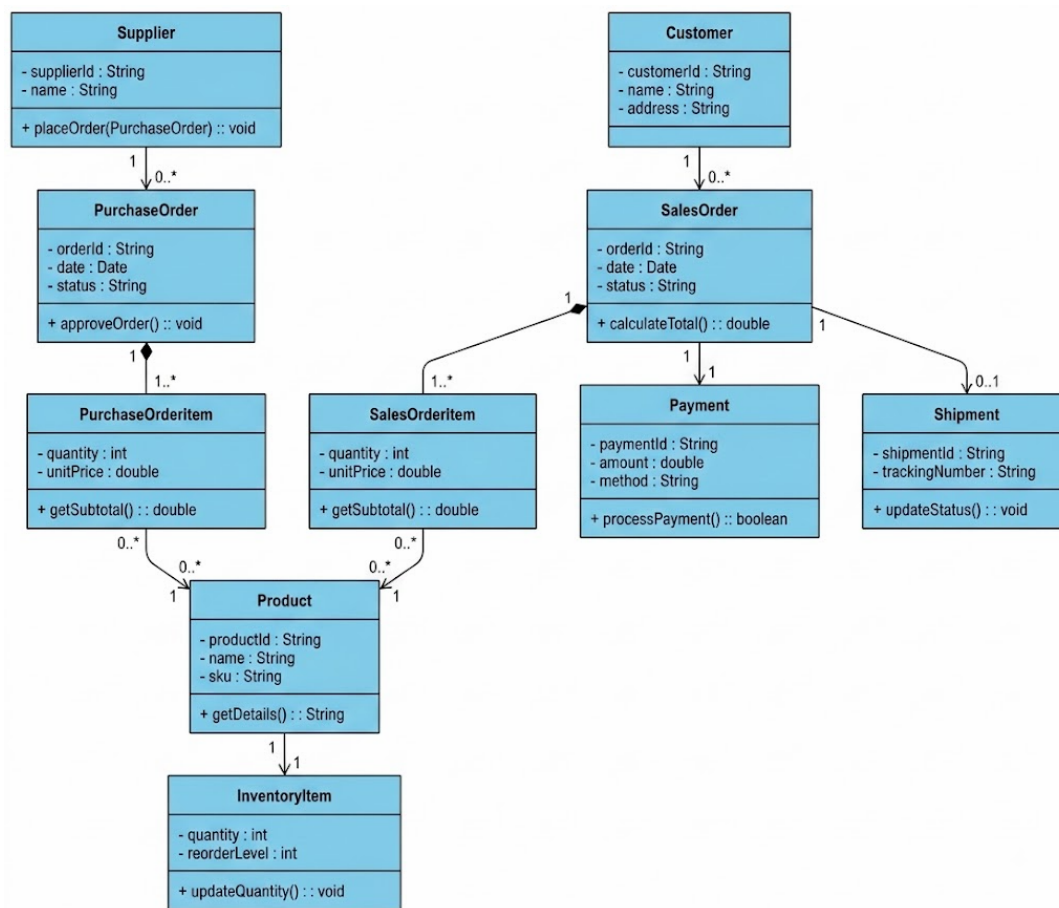


Figure 1: UML Class Diagram of Warehouse Management System

Relationship	Classes	Meaning
Association	Supplier – PurchaseOrder	A supplier can have purchase orders.
Association	Customer – SalesOrder	A customer can have sales orders.
Composition	PurchaseOrder – PurchaseOrderItem	A purchase order contains its items.
Association	SalesOrder – SalesOrderItem	A sales order is linked with its items.
Association	PurchaseOrderItem – Product	An item refers to a product.
Association	SalesOrderItem – Product	An item refers to a product.
Association	Product – InventoryItem	A product is linked with inventory.
Association	SalesOrder – Payment	A sales order is linked with payment.
Association	SalesOrder – Shipment	A sales order may have a shipment.

Multiplicity

- Supplier – PurchaseOrder: 1 to 0..* — one supplier can have multiple purchase orders.
- Customer – SalesOrder: 1 to 0..* — one customer can have multiple sales orders.
- PurchaseOrder – PurchaseOrderItem: 1 to 1..* — one purchase order contains its items.
- SalesOrder – SalesOrderItem: 1 to 1..* — one sales order has one or more items.
- PurchaseOrderItem – Product: 0..* to 1 — items can refer to one product.
- SalesOrderItem – Product: 0..* to 1 — items can refer to one product.
- Product – InventoryItem: 1 to 1 — one product is linked with one inventory item.
- SalesOrder – Payment: 1 to 1 — one sales order is linked to one payment.

- SalesOrder – Shipment: 1 to 0..1 — shipment is optional.

UML Visibility

- + Public – accessible from other classes.
- - Private – accessible only inside the class.
- # Protected – accessible by the class and its subclasses.

Conclusion

The Class Diagram gives a structural view of the Warehouse Management System. It shows the main classes, attributes, operations, relationships and multiplicities used in the system.

Result

The UML Class Diagram for the Warehouse Management System was successfully designed. The main classes and their relationships were identified and represented using UML notation.